

```
import pandas as pd

df = pd.read_csv("housing.csv")

print(df.head())

print(df.columns)

print(df.info())

print(df.isnull().sum())

print(df.shape)

# Select Features and Target
X = df[['Area', 'Bedrooms', 'Bathrooms']]
y = df['Price']

# Split Dataset
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.4,
    random_state=42
)

# Train Model
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

print("Model Trained Successfully")

# Make Predictions
y_pred = model.predict(X_test)

print("\nPredicted Prices:")
print(y_pred)

# Evaluate Model
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

print("\nModel Evaluation:")
print("MAE =", mean_absolute_error(y_test, y_pred))
print("MSE =", mean_squared_error(y_test, y_pred))
print("RMSE =", np.sqrt(mean_squared_error(y_test, y_pred)))

# R² Score may give a warning/error with very small datasets
try:
    print("R2 Score =", r2_score(y_test, y_pred))
except:
    print("R2 Score cannot be calculated with this small dataset.")

# Compare Actual vs Predicted
results = pd.DataFrame({
    'Actual Price': y_test.values,
    'Predicted Price': y_pred
})

print("\nActual vs Predicted:")
print(results)

# Predict New House Price
new_house = pd.DataFrame({
    'Area': [1600],
    'Bedrooms': [3],
    'Bathrooms': [2]
})

predicted_price = model.predict(new_house)

print("\nPredicted House Price for Area=1600, Bedrooms=3, Bathrooms=2:")
print(predicted_price[0])
```

https://colab.research.google.com/drive/1yS6Mn6jR3y2YMioAIY-ZYoZ5iYUuu8ym#scrollTo=w3o_N2XwACr1&printMode=true

```
import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()
```

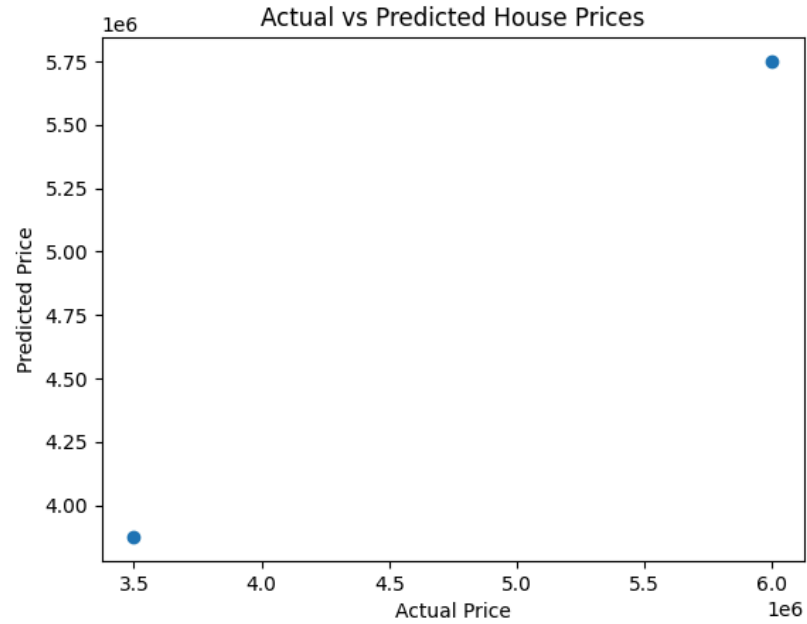
```
   Area  Bedrooms  Bathrooms   Price
0  1000         2          1  2000000
1  1500         3          2  3500000
2  1200         2          2  2500000
3  1800         4          3  5000000
4  2000         4          3  6000000
Index(['Area', 'Bedrooms', 'Bathrooms', 'Price'], dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Area        5 non-null      int64
1  Bedrooms    5 non-null      int64
2  Bathrooms    5 non-null      int64
3    Price       5 non-null      int64
dtypes: int64(4)
memory usage: 292.0 bytes
None
Area          0
Bedrooms      0
Bathrooms     0
Price         0
dtype: int64
(5, 4)
Model Trained Successfully

Predicted Prices:
[3874995.31255859 5749990.62511719]
```

```
Model Evaluation:
MAE = 312502.3437207027
MSE = 101563085985.10583
RMSE = 318689.6389672966
R2 Score = 0.9349996249695323
```

```
Actual vs Predicted:
   Actual Price  Predicted Price
0    3500000    3.874995e+06
1    6000000    5.749991e+06
```

Predicted House Price for Area=1600, Bedrooms=3, Bathrooms=2:
4249990.625117186



Start coding or [generate](#) with AI.

